SYLLABUS : CALORIMETRY AND THERMAL EXPANSION

1. A small quantity, mass m, of water at a temperature θ (in°C) is poured on to a large mass M of ice which is at its melting point. If c is the specific heat capacity of water and L the latent heat of fusion of ice, then the mass of ice melted is given by :

(A) $\frac{ML}{mc\theta}$ (B) $\frac{mc\theta}{ML}$ (C) $\frac{Mc\theta}{L}$ (D) $\frac{mc\theta}{L}$

2. A thermally isolated vessel contains 100 g of water at 0°C. When air above the water is pumped out, some of the water freezes and some evaporates at 0°C itself. then the mass of the ice formed if no water is left in the vessel. Latent heat of vaporization of water at 0°C = 2.10×10^6 J/kg and latent heat of fusion of ice = 3.36×10^5 J/kg.

(A) 86.2 g (B) 13.8 g (C) 76.2 g (D) 65.6 g

3. 20 gm ice at –10 °C is mixed with m gm steam at 100 °C. Minimum value of m so that finally all ice and steam converts into water. (Use s_{ice}=0.5 cal/gm°C,s_{water}=1 cal/gm°C,L (melting) = 80 cal/gm and L (vaporization) = 540 cal/gm)

(A) $\frac{85}{32}$ gm (B) $\frac{85}{64}$ gm (C) $\frac{32}{85}$ gm (D) $\frac{64}{85}$ gm

4. 2 kg ice at – 20 °C is mixed with 5 kg water at 20 °C . Then final amount of water in the mixture will be : [Specific heat of ice = 0.5 cal/gm °C, Specific heat of water = 1 cal/gm °C, Latent heat of fusion of ice = 80 cal/gm]

(A) 6 kg (B) 7 kg (C) 3.5 kg (D) 5 kg

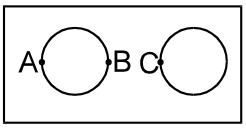
5. From what height should a piece of ice (0°C) fall so that it melts completely? Only one-quarter of the energy produced is absorbed by the ice as heat. (Latent heat of ice = 3.4×10^5 J kg^{-1}, g = 10 m/s^2)

(A) 60 km (B) 90 km (C) 125 km (D) 136 km

6. 300 g of water at 25°C is added to 100 g of ice at 0°C. Find the final temperature of the mixture.

(A) 0°C (B) 1°C (C) 2°C (D) 2°C

7. Two large holes are cut in a metal sheet. If this is heated, distances AB and BC, (as shown)



(A) both will increase (B) both will decrease

(C) AB increases, BC decreases (D) AB decreases, BC increases

8. A steel scale is to be prepared such that the millimeter intervals are to be accurate within 6×10^{-5} mm. The maximum temperature variation from the temperature of calibration during the reading of the millimeter marks is ($\alpha = 12 \times 10^{-6}$ /ºC)

(A) 4.0°C (B) 4.5°C (C) 5.0°C (D) 5.5°C

9. Expansion during heating –

(A) occurs only in a solid

(B) increases the density of the material

(C) decreases the density of the material

(D) occurs at the same rate for all liquids and solids.

10. If a bimetallic strip is heated, it will

(A) bend towards the metal with lower thermal expansion coefficient.

(B) bend towards the metal with higher thermal expansion coefficient.

(C) twist itself into helix.

(D) have no bending.

11. Two rods, one of aluminium and the other made of steel, having initial length $\ell_1$and ℓ_2 are connected together to form a single rod of length $\ell_1 + \ell_2$. The coefficients of linear expansion for aluminium and steel are α_a and α_s respectively. If the length of each rod increases by the same amount when their temperature are raised by tºC, then find the ratio $\frac{\ell_1}{(\ell_1 + \ell_2)}$.

(A) $\frac{\alpha_s}{\alpha_a}$ (B) $\frac{\alpha_a}{\alpha_s}$ (C) $\frac{\alpha_s}{(\alpha_a + \alpha_s)}$ (D) $\frac{\alpha_a}{(\alpha_a + \alpha_s)}$

12. A liquid with coefficient of volume expansion γ is filled in a container of a material having the coefficient of linear expansion α .If the liquid overflows on heating, then –

(A) $\gamma > 3\alpha$ (B) $\gamma < 3\alpha$ (C) $\gamma = 3\alpha$ (D) none of these

13. What should be the sum of lengths of an aluminium and steel rod at 0ºC is, so that at all temperatures their difference in length is 0.25m. (Take coefficient of linear expansion for aluminium and steel at 0ºC as 22×10^{-6} / ºC and 11×10^{-6} / ºC respectively.)

(A) 0.25m (B) 0.50m (C) 0.75m (D) 0.90m

14. A steel tape is correctly calibrated at 20 ºC and is used to measure the length of a table at 30 ºC. Find the percentage error in the measurement of length. [$\alpha_{steel} = 11 \times 10^{-6}$/ ºC]

(A) 1.1×10^2 (B) 1.1×10^{-2} (C) 2.1×10^{-2} (D) 1.1×10^{-3}

15. A difference of temperature of 25º C is equivalent to a difference of :

(A) 45º F (B) 72º F (C) 32º F (D) 25º F

16. A metal ball of specific gravity 4.5 and specific heat 0.1 cal/gm-°C is placed on a large slab of ice at 0°C. When ball's temperature become 0°C then half of the ball sinks in the ice. The initial temperature of the ball is :-

(Latent heat capacity of ice = 80 cal/g, specific gravity of ice = 0.9)

(A) 100 °C (B) 90 °C (C) 80 °C (D) 70 °C

17. In a steel factory it is found that to maintain M kg of iron in the molten state at its melting point an input power P watt is required. When the power source is turned off, the sample completely solidifies in time t second. The latent heat of fusion of iron is.

(A) 2 Pt / M (B) Pt / 2M (C) Pt / M (D) PM / t

18. Steam at 100º C is passed into 1.1 kg of water contained in a calorimeter of water equivalent 0.02 kg at 15º C till the temperature of the calorimeter and its contents rises to 80º C. The mass of the steam condensed in kilogram is :

(A) 0.130 (B) 0.065 (C) 0.260 (D) 0.135

19. If I is the moment of inertia of a solid body having α-coefficient of linear expansion then the change in I corresponding to a small change in temperature ΔT is

(A) $\alpha I \Delta T$ (B) $\frac{1}{2}\alpha I \Delta T$ (C) $2\alpha I \Delta T$ (D) $3\alpha I \Delta T$

20. Two rods having length ℓ_1 and ℓ_2, made of materials with the linear coefficient of expansion α_1 and α_2, were welded together. The equivalent coefficients of linear expansion for the obtained rod :-



(A) $\frac{\ell_1\alpha_2 + \ell_2\alpha_1}{\ell_1 + \ell_2}$ (B) $\frac{\ell_1\alpha_1 + \ell_2\alpha_2}{\alpha_1 + \alpha_2}$ (C) $\frac{\ell_1\alpha_1 + \ell_2\alpha_2}{\ell_1 + \ell_2}$ (D) $\frac{\ell_2\alpha_1 + \ell_1\alpha_2}{\alpha_1 + \alpha_2}$

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ANSWER KEY

1.	(B)	**2.**	(A)	**3.**	(A)	**4.**	(A)	**5.**	(D)
6.	(A)	**7.**	(A)	**8.**	(C)	**9.**	(C)	**10.**	(A)
11.	(C)	**12.**	(A)	**13.**	(C)	**14.**	(B)	**15.**	(A)
16.	(C)	**17.**	(C)	**18.**	(A)	**19.**	(C)	**20.**	(C)